Physics Practice Examination
Answer Key and Worked Solutions

Problem 1 [FY1]

Theory gives $T = 2\pi\sqrt{(l+r)/g}$, hence

$$T^2 = \frac{4\pi^2}{g}\,l + \frac{4\pi^2 r}{g},$$

a straight line in l with slope $k = 4\pi^2/g$ and intercept $b = 4\pi^2 r/g$. It must be T^2 that is plotted: T against l is a square-root curve, and fitting it with a straight line would be a modelling error. The intercept is nonzero precisely because of the bob radius, so the model must keep r.

(a) As above: $k = 4\pi^2/g$, $b = 4\pi^2 r/g$.

(b) Least squares (unweighted; no uncertainties are quoted):

$$T^2 = 4.0805\,l + 0.02242 \qquad (T^2 \text{ in } \mathrm{s}^2,\ l \text{ in m}),$$

so

$$g = \frac{4\pi^2}{4.0805\,\mathrm{s}^2/\mathrm{m}} = 9.67\,\mathrm{m/s}^2.$$

Residuals in T^2 are -3.1, 1.7, 3.6, 0.1, $-2.3\,\mathrm{ms}^2$, giving $s_y = 3.2\cdot10^{-3}\,\mathrm{s}^2$ with $\nu = 3$ degrees of freedom and

$$s_k = \frac{s_y}{\sqrt{\sum(l_i-\bar{l})^2}} = 5.1\cdot10^{-3}\,\mathrm{s}^2/\mathrm{m}, \qquad \frac{s_k}{k} = 0.12\%.$$

Since $g \propto k^{-1}$, $\Delta g/g = \Delta k/k$, so

$$\boxed{g = (9.67 \pm 0.01)\,\mathrm{m/s}^2} \qquad (0.12\%).$$

This is a *statistical* uncertainty: it describes the scatter of these five points only.

(c) Intercept $b = (4\pi^2/g)r$, with standard uncertainty $s_b = s_y\sqrt{1/n + \bar{l}^2/\sum(l_i-\bar{l})^2} = 3.4\cdot10^{-3}\,\mathrm{s}^2$. Then

$$r = \frac{b\,g}{4\pi^2} = 5.5\,\mathrm{mm}, \qquad \Delta r \approx 0.8\,\mathrm{mm},$$

and $b/s_b = 6.6$. The intercept is significantly positive (about 6σ), consistent with a bob of roughly 1 cm diameter. Fitting a model with $r = 0$ imposed would be a modelling error.

(d) The finite-amplitude correction is

$$T = T_0\Big(1 + \frac{\theta_0^2}{16}\Big), \qquad \theta_0 = 20.0^\circ = 0.3491\,\mathrm{rad}, \quad \frac{\theta_0^2}{16} = 0.00762.$$

Since $g \propto 1/T^2$, the fitted slope is too large by $(1+0.00762)^2 = 1.01529$, so

$$g_{\mathrm{wrong}} = \frac{9.81}{1.01529} = 9.663\,\mathrm{m/s}^2,$$

i.e. 1.50% low, against the observed 1.4%. The correction therefore accounts for the discrepancy, and applying it gives

$$g_{\mathrm{corr}} = 9.67\cdot1.01529 = \boxed{9.82\,\mathrm{m/s}^2}.$$

This is a *systematic* error: it shifts g in one direction, it is *not* reduced by more repetitions, and it is invisible in the fit residuals — the points lie on a perfect straight line with the wrong slope. The statistical uncertainty from part (b) therefore does not cover the deviation, which is why the result looked inconsistent. Removing it:

- Vary the amplitude (5°, 10°, 20°) and extrapolate T to $\theta_0 \to 0$. This is the correct procedure.
- Work at such small amplitudes that the correction is below the timing resolution.
- With the photogate, time *many* periods and divide.

The reading uncertainty $\pm 0.5°$ alone gives only about $\pm 0.1\%$ in g (differentiate $(1+\theta_0^2/16)^2$), so this is not a reading-uncertainty problem at all: it is a model error.

Problem 2 [FY2]

(a) The kinetic energy of the air of mass m passing the disc in time Δt is $\frac{1}{2}mv^2$ with $m = \rho A v \Delta t$; hence $P = \frac{1}{2}\rho A v^2 \cdot v = \frac{1}{2}\rho A v^3$. Dimensional check: $\mathrm{kg/m^3 \cdot m^2 \cdot m^3/s^3 = J/s}$. With $A = \pi D^2/4 = 1.131 \cdot 10^4\,\mathrm{m^2}$ and $v = 12\,\mathrm{m/s}$,

$$P_w = 1.17 \cdot 10^7\,\mathrm{W} = \boxed{11.7\,\mathrm{MW}}.$$

Air cannot be stopped dead: mass conservation forces it to leave the rotor with some speed, so the extracted power is $P = \frac{1}{2}\dot{m}(v_1^2 - v_2^2)$ with $v_2 > 0$. Optimising over v_2 gives the Betz limit $C_{p,\max} = 16/27 = 0.593$. At 12 m/s the turbine delivers 2.50 MW, i.e. 21% of P_w and 36% of the Betz limit; it is well below Betz because 12 m/s is already at the rated point, where the controller deliberately spills power.

(b) With $\bar{v} = 7.5\,\mathrm{m/s}$, $\frac{1}{2}\rho A \bar{v}^3 = 2.863\,\mathrm{MW}$, and

$$\overline{v^3} = \frac{6}{\pi}\bar{v}^3 = 1.9099\,\bar{v}^3 \implies \overline{P}_{\mathrm{avail}} = \tfrac{1}{2}\rho A \overline{v^3} = 5.468\,\mathrm{MW}.$$

Hence

$$C_p = \frac{2.50\,\mathrm{MW}}{5.468\,\mathrm{MW}} = \boxed{0.457} < 0.593.$$

Using $\bar{v}^3$ would give $C_p = 0.87 > 16/27$, which is impossible. Because $v \mapsto v^3$ is convex, $\overline{v^3} \geq \bar{v}^3$, so substituting the mean speed *underestimates* the available power and *overestimates* C_p. The two coincide only for a perfectly steady wind.

(c) $E = P_{\mathrm{rated}}\, t\, \mathrm{CF} = 2.50\,\mathrm{MW} \cdot 8760\,\mathrm{h} \cdot 0.35 = 7.67\,\mathrm{GWh}$, so

$$N = \frac{0.010 \cdot 90\,\mathrm{TWh}}{7.67\,\mathrm{GWh}} \approx \boxed{1.2 \cdot 10^2\ \text{turbines}}.$$

CO_2 avoided: $7.665 \cdot 10^6\,\mathrm{kWh} \cdot 0.90\,\mathrm{kg/kWh} = 6.9 \cdot 10^6\,\mathrm{kg} \approx \boxed{6.9 \cdot 10^3\,\mathrm{t}}$ per year.

(d) $\overline{v^3} = \frac{1}{2}(5^3 + 10^3) = 562.5$ whereas $\bar{v}^3 = 7.5^3 = 421.9$, so the alternating site yields

$$\frac{562.5}{421.875} = \boxed{\frac{4}{3}}$$

times more power — a 33% gain from the same mean speed. This is why the average power involves $\overline{v^3}$. A real turbine does not keep producing more at ever higher wind speed because it is rated at a design wind speed ($\approx$ 12–13 m/s here): above it, pitch control sheds power to protect the structure, and at storm speeds the machine is shut down.

Problem 3 [FY3]

(a) $\rho_0 = p_0 M/(RT_0) = 101.3 \cdot 10^3 \cdot 0.0290/(8.314 \cdot 293.15) = \boxed{1.205\,\mathrm{kg/m^3}}$.

(b) Buoyancy must balance the total weight,

$$V\rho_0 g = V\rho_i g + (m_{\mathrm{env}} + m_{\mathrm{bask}} + m_{\mathrm{pay}})g,$$

so the required internal density is

$$\rho_i = \rho_0 - \frac{m_{\text{tot}}}{V} = 1.205 - \frac{180 + 90 + 350}{2.2 \cdot 10^3} = 1.205 - 0.2818 = 0.9235\,\text{kg/m}^3.$$

At constant pressure $\rho \propto 1/T$, hence

$$T_i = T_0 \frac{\rho_0}{\rho_i} = 293.15 \cdot \frac{1.2053}{0.9235} = 382.6\,\text{K} = \boxed{109^\circ\text{C}}.$$

(c) At $T_i = 393.15\,\text{K}$, $\rho_i = \rho_0 T_0 / T_i = 0.8991\,\text{kg/m}^3$, so the total mass that can be carried is

$$m_{\text{tot}} = V(\rho_0 - \rho_i) = 2.2 \cdot 10^3 \cdot 0.3063 = 674\,\text{kg},$$

leaving a payload of $674 - 270 = \boxed{4.0 \cdot 10^2\,\text{kg}}$.

(d) The burner must supply the enthalpy to raise the entering air from T_0 to T_i:

$$\dot{m} = \rho_i \dot{V} = 0.9235 \cdot 0.45 = 0.4156\,\text{kg/s},$$

$$P = \dot{m}\, c_p\, (T_i - T_0) = 0.4156 \cdot 1005 \cdot (382.6 - 293.15) = \boxed{3.7 \cdot 10^4\,\text{W} \approx 37\,\text{kW}}.$$

Approximations: (i) constant pressure, so c_p is the right specific heat; (ii) the air leaves at the same temperature at which it enters, i.e. no heat is lost through the fabric and the mixing/leakage losses are ignored. Every one of these makes the estimate a *lower bound*: real envelopes lose heat and the expelled air carries away more enthalpy than assumed. That 37 kW is unrealistically small for a $2200\,\text{m}^3$ balloon (typical burners are hundreds of kW) is exactly the size of the neglected losses.

Problem 4 [FY4]

(a) With B descending and A moving up the plane, the driving force is

$$m_B g - m_A g \sin\theta = 29.43 - 19.62 = 9.81\,\text{N},$$

while the maximum static friction is $\mu_s m_A g \cos\theta = 0.25 \cdot 33.98 = 8.50\,\text{N}$. Since $9.81 > 8.50$, A does move as assumed. Then $f_k = \mu_k m_A g \cos\theta = 6.80\,\text{N}$ and

$$a = \frac{m_B g - m_A g \sin\theta - f_k}{m_A + m_B} = \frac{9.81 - 6.80}{7.00} = \boxed{0.430\,\text{m/s}^2},$$

$$T = m_B(g - a) = 3.00(9.81 - 0.430) = \boxed{28.1\,\text{N}}.$$

Check on A: $T - m_A g \sin\theta - f_k = 28.14 - 19.62 - 6.80 = 1.72\,\text{N}$ and $m_A a = 1.72\,\text{N}$. ✓

(b) $v = \sqrt{2ah} = \sqrt{2 \cdot 0.4305 \cdot 0.80} = \boxed{0.830\,\text{m/s}}$. Energy balance (all terms in J):

$$\underbrace{m_B g h}_{23.54} = \underbrace{m_A g h \sin\theta}_{\Delta U_A = 15.70} + \underbrace{f_k h}_{W_f = 5.44} + \underbrace{\tfrac{1}{2}(m_A + m_B)v^2}_{K = 2.41},$$

and $15.70 + 5.44 + 2.41 = 23.54$. ✓

(c) After B lands, only gravity and friction act on A:

$$a_2 = g(\sin\theta + \mu_k \cos\theta) = 9.81(0.500 + 0.1732) = 6.60\,\text{m/s}^2$$

(deceleration), so

$$s = \frac{v^2}{2a_2} = \frac{0.6887}{13.21} = \boxed{5.2\,\text{cm}}.$$

A stays put only if $\mu_s \geq \tan\theta = 0.577$; here $\mu_s = 0.25$, so A slides back down.

(d) Sliding down, $a_3 = g(\sin\theta - \mu_k\cos\theta) = 9.81(0.500 - 0.1732) = 3.21\,\mathrm{m/s^2}$, and the distance back to the point where B landed is $s = 5.2\,\mathrm{cm}$. So A's speed there is

$$v_A = \sqrt{2a_3 s} = \sqrt{2 \cdot 3.206 \cdot 0.05215} = \boxed{0.578\,\mathrm{m/s}}.$$

At that instant the string snaps taut. The jerk is so brief that the impulses of gravity and friction are negligible; the tension impulse $J = \int T\,dt$ is internal to the pair A+B, so momentum is conserved. B is at rest on the floor at that moment and is jerked into motion, so the two blocks end up with a common speed

$$(m_A + m_B)V = m_A v_A \implies V = \frac{m_A}{m_A + m_B} v_A = \frac{4}{7} \cdot 0.578 = \boxed{0.33\,\mathrm{m/s}},$$

with an impulse $J = m_B V = \boxed{0.99\,\mathrm{N\,s}}$ transferred from A to B (the same J equals $m_A(v_A - V)$). The kinetic energy is

$$K_{\text{before}} = \tfrac{1}{2} m_A v_A^2 = 0.669\,\mathrm{J}, \qquad K_{\text{after}} = \tfrac{1}{2}(m_A + m_B)V^2 = 0.382\,\mathrm{J},$$

so $0.287\,\mathrm{J}$ is lost in the inelastic jerk, i.e. the fraction

$$\frac{|\Delta K|}{K} = \frac{m_B}{m_A + m_B} = \frac{3}{7} = 0.43,$$

exactly as in a perfectly inelastic collision, and this fraction does not depend on the masses' arrangement.

What happens next is a nice check on the sign conventions: A is moving down the plane and B upward, so gravity now opposes the motion, and with the string taut

$$a = \frac{m_A g\sin\theta - m_B g - \mu_k m_A g\cos\theta}{m_A + m_B} = \frac{19.62 - 29.43 - 6.80}{7.00} = -2.4\,\mathrm{m/s^2},$$

i.e. the pair decelerates. Starting from $V = 0.33\,\mathrm{m/s}$ it stops after only $0.023\,\mathrm{m}$ and $0.14\,\mathrm{s}$, whereupon B falls back onto the floor and the string goes slack again. The blocks therefore descend in a sequence of small jerks, losing energy at every one; the total energy lost is not recoverable, which is why a taut inelastic string is not a conservative constraint.

Problem 5 [FY5]

(a) $r = R_E + h = 6.771 \cdot 10^6\,\mathrm{m}$ and $GM_E = 3.984 \cdot 10^{14}\,\mathrm{m^3/s^2}$.

$$v = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{3.984 \cdot 10^{14}}{6.771 \cdot 10^6}} = \boxed{7.67\,\mathrm{km/s}},$$

$$T = \frac{2\pi r}{v} = 5.55 \cdot 10^3\,\mathrm{s} = \boxed{92.4\,\mathrm{min}},$$

$$g(r) = \frac{GM_E}{r^2} = \boxed{8.69\,\mathrm{m/s^2}}.$$

(b) Circular motion requires $mv^2/r = GM_E m/r^2$, hence $K = \tfrac{1}{2}mv^2 = \tfrac{1}{2}GM_E m/r$. Since $U = -GM_E m/r$, we get $K = -\tfrac{1}{2}U$ and

$$E = K + U = \tfrac{1}{2}U = -\frac{GM_E m}{2r}.$$

With $m = 4.2 \cdot 10^5\,\mathrm{kg}$,

$$E = -\frac{3.984 \cdot 10^{14} \cdot 4.2 \cdot 10^5}{2 \cdot 6.771 \cdot 10^6} = -1.24 \cdot 10^{13}\,\mathrm{J} = \boxed{-12.4\,\mathrm{TJ}}.$$

Escape speed: $v_{\text{esc}} = \sqrt{2GM_E/r} = 10.8\,\text{km/s} = \sqrt{2}\,v$. The two numbers differ by exactly this factor because escaping means supplying $|E| = K$: you need to raise E from $-K$ to 0.

(c) $E \propto -1/r$, so with $\Delta r = -1.00\,\text{km}$,

$$\Delta E \approx -\frac{|E|\,\Delta r}{r} = -\frac{1.236 \cdot 10^{13} \cdot (-10^3)}{6.771 \cdot 10^6} = \boxed{+1.8\,\text{GJ}}.$$

The total energy *decreases* by 1.8 GJ: that energy must be removed from the orbit (by drag or by a retro burn). Of it, the potential energy falls by $2 \times 1.8\,\text{GJ} = 3.7\,\text{GJ}$ while the kinetic energy rises by 1.8 GJ (since $K = -E$). From $v = \sqrt{GM/r}$, $\mathrm{d}v = -\frac{1}{2}(v/r)\mathrm{d}r$, so

$$\Delta v = +\tfrac{1}{2}\frac{v}{r} \cdot 10^3 = \boxed{+0.57\,\text{m/s}},$$

and $T \propto r^{3/2}$ gives $\Delta T = \frac{3}{2}(T/r)\Delta r = -1.2\,\text{s}$. So the station speeds *up* by 0.57 m/s while losing energy: a retro burn lowers the orbit and converts more potential energy into kinetic energy than the burn removes. A prograde burn instead raises the orbit and slows the station. This is the reverse of the intuition from driving a car.

(d) $a = \omega^2 R_s = 0.256^2 \cdot 50.0 = 3.28\,\text{m/s}^2$. At the head, $R = 50.0 - 1.80 = 48.2\,\text{m}$, so

$$a_{\text{head}} = \omega^2 \cdot 48.2 = 3.16\,\text{m/s}^2,$$

i.e. $1 - 48.2/50.0 = 3.6\%$ weaker. The relative gradient is $\Delta a/a = \Delta R/R_s$, so to keep it below a few per cent R_s must be large compared with human height: a healthy habitat needs a radius of tens or hundreds of metres. Small habitats suffer Coriolis artefacts, gradients and vertigo.

(e) In the habitat frame the pendulum's equilibrium direction is radial and the restoring "gravity" is the centrifugal acceleration at its own location; taking the bob's radius as $\approx 50\,\text{m}$, so that $g_{\text{eff}} = a = 3.28\,\text{m/s}^2$,

$$T = 2\pi\sqrt{\frac{L}{a}} = 2\pi\sqrt{\frac{3.00}{3.28}} = \boxed{6.0\,\text{s}},$$

against $2\pi\sqrt{3.00/9.81} = 3.5\,\text{s}$ on Earth: longer by $\sqrt{9.81/3.28} = \sqrt{3}$, since the period scales as $g^{-1/2}$. (Strictly the bob sits 3.00 m inboard of the feet, where a is 6% smaller; using $R = 47$ and 50 m gives 6.2 and 6.0 s respectively. The point of the question is the scaling, not the third digit.)

Problem 6 [FY5]

(a) $s = A\sin\omega t$ with $\omega = 2\pi f = 2.765 \cdot 10^3\,\text{rad/s}$ and $A = 5.00 \cdot 10^{-4}\,\text{m}$:

$$v_{\max} = \omega A = \boxed{1.38\,\text{m/s}}, \qquad a_{\max} = \omega^2 A = \boxed{3.82 \cdot 10^3\,\text{m/s}^2} \approx 390\,g.$$

The air molecules oscillate about fixed positions; they do not travel with the wave. For a pipe closed at one end, resonances occur at $l_n + \Delta l = (2n-1)\lambda/4$, so

$$l_2 - l_1 = \frac{\lambda}{2} \implies \lambda = 2(0.572 - 0.182) = 0.780\,\text{m},$$

$$v = f\lambda = 440 \cdot 0.780 = \boxed{343\,\text{m/s}}.$$

(b) The ideal quarter-wave length is $\lambda/4 = 0.195\,\text{m}$, but the resonance is at 0.182 m: the air just outside the open end also moves, effectively lengthening the pipe. In this convention $l_{\text{resonance}} + \Delta l = \lambda/4$, so

$$\Delta l = \frac{\lambda}{4} - l_1 = 0.195 - 0.182 = 1.3\,\text{cm}.$$

With $\Delta l = 0.6r$, $r = 1.3/0.6 = 2.2\,\text{cm}$ (diameter about 4.3 cm) — a plausible resonance tube. Δl cancels in $l_2 - l_1$ because it is the same for every resonance, but it does not cancel in Δl itself, which is why the absolute positions are what determine the tube geometry.

(c) $f_2 = f \pm 4.0\,\text{Hz}$, i.e. 436 or 444 Hz. To decide: load the unknown fork slightly (a dab of plasticine). Loading lowers its frequency, so the beat frequency increases if f_2 was already above 440 Hz, and decreases if it was below. Beats occur at $|f_1 - f_2| = 4.0$ per second, so in 5.0 s there are $\boxed{20}$ beats.

(d) With the same end correction, $\lambda_{CO_2} = 4(l_1 + \Delta l) = 4(0.140 + 0.013) = 0.612\,\text{m}$, so

$$v_{CO_2} = 440 \cdot 0.612 = \boxed{269\,\text{m/s}}.$$

Kinetic theory gives $v = \sqrt{\gamma RT/M}$:

$$v_{\text{air}} = \sqrt{\frac{1.40 \cdot 8.314 \cdot 293}{0.0290}} = 343\,\text{m/s}, \qquad v_{CO_2} = \sqrt{\frac{1.30 \cdot 8.314 \cdot 293}{0.0440}} = 268\,\text{m/s}.$$

The measured value agrees within 0.4%. The speed drops because the larger molar mass dominates the slightly smaller γ: $v \propto \sqrt{\gamma/M}$, and M rises by a factor 1.52 while γ falls by only 7%. The beat frequency of part (c) is a property of the two forks, not of the medium, so it is unchanged; only the wavelength changes.

Problem 7 [FY6]

(a) $R_1 \parallel R_2$: $R_p = 18 \cdot 22/40 = 9.90\,\Omega$. Total $R_{\text{tot}} = 9.90 + 15.0 + 3.0 = 27.9\,\Omega$, so

$$I = \frac{12.0}{27.9} = \boxed{0.430\,\text{A}}.$$

Terminal voltage $U = E - Ir = 12.0 - 1.29 = \boxed{10.7\,\text{V}}$. The parallel voltage is $U_p = IR_p = 4.258\,\text{V}$, hence

$$I_{R_1} = \frac{4.258}{18} = \boxed{0.237\,\text{A}}, \qquad I_{R_2} = \frac{4.258}{22} = \boxed{0.194\,\text{A}},$$

and $0.237 + 0.194 = 0.430\,\text{A}$. ✓ The capacitor carries U_p:

$$Q = CU_p = 2.2 \cdot 10^{-6} \cdot 4.258 = \boxed{9.4\,\mu\text{C}}, \qquad W = \tfrac{1}{2}CU_p^2 = \boxed{2.0 \cdot 10^{-5}\,\text{J}}.$$

(b) Powers:

$$R_3 : I^2R_3 = 2.77\,\text{W} \qquad R_1 : U_p^2/R_1 = 1.01\,\text{W}$$
$$R_2 : U_p^2/R_2 = 0.824\,\text{W} \qquad r : I^2r = 0.555\,\text{W}$$

summing to 5.16 W, while $EI = 12.0 \cdot 0.430 = 5.16\,\text{W}$. ✓ Efficiency (external resistors only):

$$\eta = \frac{2.77 + 1.01 + 0.824}{5.16} = 0.892 = \boxed{89\%}.$$

(c) With the voltmeter across the parallel pair, R_1 is loaded:

$$R_1' = \frac{R_1 R_V}{R_1 + R_V}, \quad R_p' = \frac{R_1' R_2}{R_1' + R_2}, \quad I' = \frac{E}{R_p' + R_3 + r}, \quad U' = I' R_p'.$$

As $R_V \to \infty$, $U = 4.258\,\text{V}$. $R_V = 200\,\Omega$: $R_1' = 16.51$, $R_p' = 9.45\,\Omega$, $I' = 0.437\,\text{A}$, $U' = \boxed{4.13\,\text{V}}$ (3% low). $R_V = 30\,\Omega$: $R_1' = 11.25$, $R_p' = 7.44\,\Omega$, $I' = 0.471\,\text{A}$, $U' = \boxed{3.51\,\text{V}}$ (18% low). The reading is always too small because the voltmeter draws current, lowering the equivalent resistance of the pair and therefore its voltage drop. Solving $U'(R_V) = 0.99\,U$ gives $R_V \approx 6.3 \cdot 10^2\,\Omega$, i.e. some 60× the 9.9 Ω it is in parallel with. The practical rule is $R_V \gg R_{\text{measured}}$; real meters have MΩ inputs, so the error vanishes.

(d) A diode's I–U characteristic is exponential: a small voltage change produces a huge current change, so direct connection to 9.0 V would destroy the LED. The series resistor sets the current:

$$R = \frac{U - U_f}{I} = \frac{9.0 - 2.0}{0.020} = \boxed{350\,\Omega}, \qquad P_R = I^2 R = \boxed{0.14\,\mathrm{W}}.$$

Reversed, the diode is in the blocking direction: essentially no current flows (only nA leakage), the LED does not light, and nothing is damaged. As the battery discharges its terminal voltage falls, so $U_R = U_{\mathrm{batt}} - U_f$ falls, the current $I = U_R/R$ falls and the LED dims. Because U_f is nearly pinned by the exponential characteristic, almost the whole voltage change appears across R, so the dimming is gradual.

Problem 8 [FY7]

(a) $f = 3000/60 = 50\,\mathrm{Hz}$, $\omega = 2\pi f = 314\,\mathrm{rad/s}$. With the coil normal at angle $\theta = \omega t$ to $\mathbf{B}$,

$$\Phi = BA\cos\omega t, \qquad e = -N\frac{d\Phi}{dt} = NBA\omega\sin\omega t.$$

$$\hat{e} = NBA\omega = 100 \cdot 0.85 \cdot 1.22 \cdot 10^{-2} \cdot 314 = \boxed{3.3 \cdot 10^2\,\mathrm{V}},$$

$$e_{\mathrm{rms}} = \frac{\hat{e}}{\sqrt{2}} = \boxed{2.3 \cdot 10^2\,\mathrm{V}}.$$

Over a quarter turn, $\bar{e} = N\Delta\Phi/\Delta t$ with $\Delta\Phi = BA$ and $\Delta t = T/4 = 1/(4f)$, giving

$$\bar{e} = 4NBAf = \frac{2}{\pi}\hat{e}.$$

Hence $\hat{e}/\bar{e} = \pi/2 = 1.57$. The reason is that $e(t)$ is sinusoidal: a quarter turn covers the quarter of the sine from zero to its peak, whose mean is $(2/\pi)$ times the peak, just as for any sinusoid. (The mean of $|e|$ over a full period is also $(2/\pi)\hat{e}$, while the rms is $\hat{e}/\sqrt{2}$; these three averages must not be confused.)

(b) Without a transformer, at $U = 230\,\mathrm{V}$,

$$I = \frac{P}{U} = \frac{50 \cdot 10^6}{230} = 2.2 \cdot 10^5\,\mathrm{A}, \qquad P_{\mathrm{loss}} = I^2 R = \boxed{4.7 \cdot 10^{11}\,\mathrm{W}},$$

which is nearly 10^4 times the generated power — utterly impossible. That is itself the answer: without a transformer the line cannot deliver the power at all. In general

$$P_{\mathrm{loss}} = I^2 R = \frac{P^2 R}{U^2},$$

so the loss scales as U^{-2}. With $U' = 110\,\mathrm{kV}$: $I = 455\,\mathrm{A}$ and

$$P_{\mathrm{loss}} = (455)^2 \cdot 10 = \boxed{2.1 \cdot 10^6\,\mathrm{W} = 2.1\,\mathrm{MW}},$$

which is $\boxed{4.1\%}$ of 50 MW. A transformer needs a *changing* flux: for DC, $\Phi \propto I$ is constant, so the secondary emf $N_2\,d\Phi/dt = 0$. DC cannot be transformed, which is why HVDC links convert to AC or use power electronics at both ends. Raising the voltage is a cost trade: line loss falls as U^{-2} while insulation, tower height and clearances cost grows with U. The optimum balances the two. It is not about transformer efficiency, which already exceeds 99%.

Problem 9 [FY7]

(a) Let the rod slide down the incline at speed v. The field is perpendicular to the rail plane, so $\mathcal{E} = BLv$, $I = BLv/R$, and the magnetic force on the rod is $F = BIL = B^2L^2v/R$, directed *up* the incline.

i. Terminal speed follows from $mg\sin\theta = B^2L^2v_t/R$:

$$v_t = \frac{mgR\sin\theta}{B^2L^2} = \frac{0.20\cdot 9.81\cdot 0.30\cdot 0.500}{1.2^2\cdot 0.30^2} = \boxed{2.27\,\text{m/s}}.$$

ii. $I = BLv_t/R = 1.2\cdot 0.30\cdot 2.27/0.30 = \boxed{2.73\,\text{A}}$. Electrical power

$$P_e = I^2R = 2.73^2\cdot 0.30 = \boxed{2.23\,\text{W}},$$

while the mechanical power released is $mg\sin\theta\, v_t = 0.981\cdot 2.27 = 2.23\,\text{W}$. ✓

iii. Pushed up at 2.0 m/s, the motional emf is $\mathcal{E} = BLv = 1.2\cdot 0.30\cdot 2.0 = 0.72\,\text{V}$, so $I = 0.72/0.30 = \boxed{2.4\,\text{A}}$, and the magnetic force is now *down* the incline, $BIL = 1.2\cdot 2.4\cdot 0.30 = 0.864\,\text{N}$. The external force must overcome both:

$$F = mg\sin\theta + BIL = 0.981 + 0.864 = \boxed{1.85\,\text{N}}.$$

Power balance: the external power $Fv = 1.85\cdot 2.0 = 3.69\,\text{W}$ splits into $mg\sin\theta\, v = 1.96\,\text{W}$ of potential energy and $I^2R = 2.4^2\cdot 0.30 = 1.73\,\text{W}$ of Joule heat, and $1.96 + 1.73 = 3.69$. ✓ The magnetic force itself does no net work: it converts mechanical energy into electrical energy at exactly the rate it is dissipated.

iv. With $\mathbf{B}$ vertical, the flux through the loop is $BA_\perp$ with $A_\perp = A\cos\theta$, so

$$\mathcal{E} = \frac{d\Phi}{dt} = BLv\cos\theta. \qquad (*)$$

A current $I = BLv\cos\theta/R$ flows, and the magnetic force BIL is horizontal; its component along the rails is $BIL\cos\theta$, i.e.

$$F_\parallel = \frac{B^2L^2v\cos^2\theta}{R}.$$

Setting $F_\parallel = mg\sin\theta$ gives

$$v_t = \frac{mgR\sin\theta}{B^2L^2\cos^2\theta} = \frac{v_{t,\perp}}{\cos^2\theta},$$

so with $\theta = 30^\circ$ the terminal speed is larger by 4/3: $\boxed{3.03\,\text{m/s}}$. The extra factor of $1/\cos^2\theta$ has two independent origins: only the component $B\cos\theta$ perpendicular to the plane produces motional emf, and only the component $\cos\theta$ of the magnetic force lies along the rails. Both must be compensated, hence the square. (Check: with $\theta = 0$ the expression reduces to the previous case, as it must.)

(b) *i.* $\sin\theta_c = n_{cl}/n_c = 1.45/1.46 = 0.99315$, so $\theta_c = 83.3^\circ$ from the normal, i.e. 6.7° from the axis. The numerical aperture is

$$\text{NA} = \sqrt{n_c^2 - n_{cl}^2} = \sqrt{2.916 - 2.9025} = 0.171,$$

so the acceptance angle in air is $\arcsin(\text{NA}) = 9.8^\circ$. (The numerical aperture is defined for the external medium; inside the core the limiting ray makes $\arcsin(\text{NA}/n_c) = 6.7^\circ$ with the axis.)

ii. At the limiting internal angle the geometric path in 1.00 m of fibre is $1.00/\cos 6.71^\circ = 1.0069\,\text{m}$, i.e. 0.69 cm extra. The delay relative to the axial ray is

$$\Delta t = \frac{\Delta L\, n_c}{c} = \frac{0.00690\cdot 1.46}{3.00\cdot 10^8} = \boxed{3.4\cdot 10^{-11}\,\text{s/m}},$$

i.e. 34 ps per metre, or 34 ns per kilometre. Different modes therefore arrive at slightly different times, so successive pulses smear into each other (modal dispersion) and the bit rate is limited. In a

step-index fibre with a larger NA (≈ 0.3) the spread is tens of ns/km, which is severe. A graded-index core makes the longer paths travel through lower-index glass near the cladding, where the light is faster, so the transit times are approximately equalised.

iii. $A/A_0 = 10^{-\alpha d/10} = 10^{-0.25\cdot 10/10} = 10^{-0.25} = \boxed{0.56}$, so about 44% is lost over 10 km. Silica absorbs light by exciting electrons, which needs a photon energy above a threshold — in practice, in the ultraviolet. Violet photons ($h\nu \approx 2.5\,\text{eV}$) are closer to that threshold and are scattered and absorbed more strongly, while infrared photons ($\approx 0.8\,\text{eV}$) have too little energy for electronic transitions and are absorbed only by weak vibrational tails. That is why the attenuation minimum sits at 1.3–1.55 μm and why long-haul fibres use infrared.

Problem 10 [FY8]

(a) i. $E = hc/\lambda$ with $hc = 1240\,\text{eV nm}$:

$$E_1 = \frac{1240}{589.0} = 2.1050\,\text{eV}, \qquad E_2 = \frac{1240}{589.6} = 2.1029\,\text{eV},$$

so $\Delta E = 2.1\,\text{meV}$ and

$$\Delta f = \frac{\Delta E}{h} = \Delta E \cdot \frac{1\,\text{eV}}{h} = \boxed{5.2 \cdot 10^2\,\text{GHz}}.$$

(Equivalently $\Delta f \approx f\,\Delta\lambda/\lambda = 5.09 \cdot 10^{14} \cdot (0.6/589) = 5.2 \cdot 10^{11}\,\text{Hz}$.)

ii. The grating constant is $d = (1/600)\,\text{mm} = 1.667\,\mu\text{m}$, and $d\sin\theta = k\lambda$:

$$\sin\theta_1 = \frac{589.3\,\text{nm}}{1.667\,\mu\text{m}} = 0.3536 \Rightarrow \theta_1 = 20.7^\circ,$$

$$\sin\theta_2 = 0.7071 \Rightarrow \theta_2 = 45.0^\circ, \qquad \sin\theta_3 = 1.061 > 1 \text{ (no third order)}.$$

So only orders $k = 0, 1, 2$ exist. To resolve the doublet in first order, $R = \lambda/\Delta\lambda = 589.3/0.6 = 982 = kN$ with $k = 1$ requires $N = \boxed{9.8 \cdot 10^2}$ illuminated lines, i.e. about 1.6 mm of grating.

iii. Threshold wavelength $\lambda_{\max} = hc/W = 1240/2.28 = \boxed{544\,\text{nm}}$. At 500 nm,

$$E_k^{\max} = \frac{1240}{500} - 2.28 = 2.48 - 2.28 = \boxed{0.20\,\text{eV}},$$

and the stopping potential is $\boxed{0.20\,\text{V}}$. All three are *independent* of intensity; the intensity controls the *number* of photoelectrons per second (the photocurrent), not their maximum energy. This is the content of Einstein's explanation: photon energy depends only on frequency.

(b) i.

$$\Delta m = (2.014102 + 3.016049) - (4.002602 + 1.008665) = 0.018884\,\text{u},$$

$$Q = 0.018884 \cdot 931.494 = \boxed{17.59\,\text{MeV}}.$$

Taking the deuteron as the consumed fuel,

$$\frac{Q}{m_D} = \frac{17.59 \cdot 1.6022 \cdot 10^{-13}\,\text{J}}{2.014102 \cdot 1.6605 \cdot 10^{-27}\,\text{kg}} = \boxed{8.4 \cdot 10^{13}\,\text{J/kg}}.$$

Fission of U-235 at 200 MeV per nucleus gives $8.2 \cdot 10^{13}\,\text{J/kg}$ of uranium, so *per kilogram of fuel consumed* the two are within a factor 10 of each other — the familiar claim that fusion releases "millions of times" more energy is per kilogram of the light *reagents* only if one compares with chemical fuels, not with fission. The decisive advantage of fusion is not the energy per kg but the abundance of deuterium. Coal at $3.0 \cdot 10^7\,\text{J/kg}$ is a factor $2.8 \cdot 10^6$ worse: 1 kg of deuterium releases as much as about $2.8 \cdot 10^6$ kg of coal.

ii. With $P = 2.5 \cdot 10^9\,\mathrm{W}$ and $Q = 17.59\,\mathrm{MeV} = 2.818 \cdot 10^{-12}\,\mathrm{J}$,

$$\dot{N} = \frac{P}{Q} = \boxed{8.9 \cdot 10^{20}\,\mathrm{s^{-1}}}.$$

In one year ($3.156 \cdot 10^7\,\mathrm{s}$) that consumes $2.80 \cdot 10^{28}$ deuterons, i.e.

$$m_D = 2.80 \cdot 10^{28} \cdot 2.014\,\mathrm{u} = \boxed{94\,\mathrm{kg}}.$$

At 33 g of D per m^3 of seawater this is about $2.8 \cdot 10^3\,\mathrm{m}^3$ — a swimming pool per GW-plant-year. Fuel supply is emphatically *not* the limiting problem; achieving and sustaining the plasma conditions is.

(c) i. Here $t = 12.0\,\mathrm{h} = 0.500\,\mathrm{d}$, so $t/T_{1/2} = 0.500/8.02 = 0.06234$ and

$$A = A_0\,2^{-t/T_{1/2}} = 20.0 \cdot 2^{-0.06234} = \boxed{19.2\,\mathrm{GBq}},$$

a fall of only 4.2%. The number of nuclei falls by the same 4.2%, which is why it "does not visibly change": λ is tiny on this time scale. With $\lambda = \ln 2/T_{1/2} = 1.0003 \cdot 10^{-6}\,\mathrm{s^{-1}}$ and $N_0 = A_0/\lambda$,

$$N_0 = \frac{20.0 \cdot 10^9}{1.0003 \cdot 10^{-6}} = \boxed{2.0 \cdot 10^{16}}.$$

That corresponds to only about $4\,\mu\mathrm{g}$ of $^{131}\mathrm{I}$: the activity is high because the decay constant is large, not because much material is present.

ii. Imaging needs radiation that *leaves* the body. $^{131}\mathrm{I}$ is taken up by thyroid tissue, and its *gamma* rays (364 keV) penetrate tissue and reach an external camera; its eight-day half-life matches the uptake biology while decaying away afterwards. An α emitter would be stopped within a few μm of the tissue that absorbed it — no image, but a large local dose. Alpha emitters are useful for targeted *therapy*, not imaging.

(d) Gamma: $H = 0.20 \cdot 1 = \boxed{0.20\,\mathrm{Sv}}$. Alpha: $H = 0.20 \cdot 20 = \boxed{4.0\,\mathrm{Sv}}$, i.e. a potentially lethal equivalent dose, twenty times the gamma figure for the same absorbed energy. The factor $w_R = 20$ already encodes the greater biological damage of densely ionising tracks. What the grey-to-sievert conversion still ignores is the *spatial distribution* of the dose. A body-averaged 0.20 Gy from gammas is a fairly uniform exposure of all organs; the same average delivered by alphas concentrated in the lungs means a much higher local dose in a small, radio-sensitive tissue, and the relevant quantity becomes the organ dose multiplied by a tissue weighting factor w_T (≈ 0.12 for lung) rather than the whole-body average. In practice an alpha dose is always local, so quoting a body-averaged number would understate the local effect by orders of magnitude — which is exactly why inhaled alpha emitters such as radon daughters are so hazardous.